The Menger Sponge: Recursion, Self-Similarity, and Fractal Dimension in Three Dimensions

Final Project Report

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Abstract

The Menger sponge is a fractal solid built by an infinite recursive removal process. This report analyzes its structure using the discrete-mathematical tools of recurrence relations, summation, and induction. We prove closed forms for its subcube count and volume, derive an exact recurrence for its surface area and show that the surface area diverges while the volume vanishes, and establish its fractal dimension $\log 20 / \log 3$. We give two implementations, GPU ray marching with a signed-distance estimator and a sparse hierarchical voxel tree, and use them to verify the dimension empirically by box counting and to compare the two representations.

1 Introduction

The *Menger sponge* is the three-dimensional generalization of the Cantor set and the Sierpiński carpet: a solid built by an infinite recursive process that removes material at every scale. It is a striking object because it violates two intuitions at once. As the construction proceeds, its volume shrinks toward zero, yet its surface area grows without bound; and although it is built inside a solid cube, its dimension is not 3 but an irrational number strictly between 2 and 3. These are not merely visual curiosities: each follows directly from recurrence relations, geometric series, and self-similarity, the tools of Chapters 2 and 5 of the course.

This report makes the following contributions. We derive and prove by induction the closed forms for the sponge's subcube count and volume (Section 3.1); we derive an exact recurrence for its surface area and prove the volume/surface-area paradox in closed form (Section 3.3); we establish the fractal dimension both by self-similarity and by box counting (Section 3.4); and we give a base-3 characterization of membership (Section 3.5). We then describe two independent implementations, a GPU ray marcher and a sparse voxel tree (Section 4), and use them to confirm the dimension empirically and to compare the two representations quantitatively (Section 5).

2 Definitions and Background

Begin with a solid unit cube. Partition it into a $3 \times 3 \times 3$ grid of 27 congruent subcubes, and delete the 7 subcubes that touch the center (the six face-center cubes and the single central cube), leaving 20. Applying the same rule recursively to each of the 20 survivors, and continuing indefinitely, yields the Menger sponge as the limiting set. We write M_n for the level-n approximation (the solid consisting of the 20^n retained subcubes after n steps).

The sponge is *self-similar*: it is the union of 20 copies of itself, each scaled by the ratio $1/3$. In the language of iterated function systems, it is the attractor of 20 contraction maps of ratio $1/3$. This places it on a ladder of fractals generated by the same digit-deletion idea in successive dimensions:

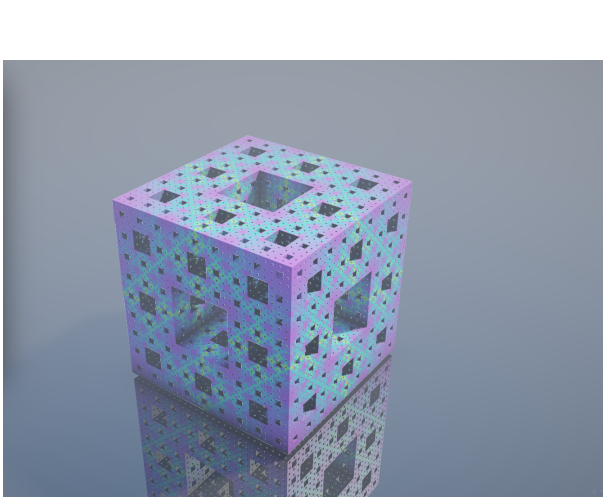
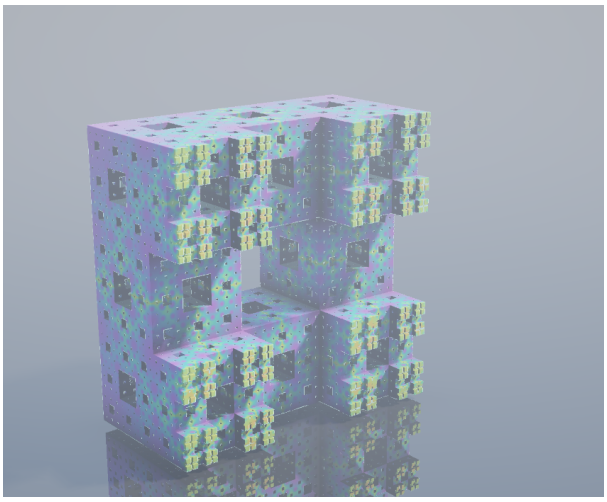



Figure 1: The Menger sponge rendered by GPU ray marching. Left: full view with reflective floor and orbit-trap surface coloring. Right: a half-space cross-section exposing the recursive internal tunnel structure; each face is a Sierpiński carpet.

the Cantor set in one dimension (keep 2 of 3), the Sierpiński carpet in two (keep 8 of 9), and the Menger sponge in three (keep 20 of 27). Indeed, every flat face of the sponge is exactly a Sierpiński carpet, a fact we use in Section 3.3.

3 Theorems

Let N_n be the number of retained subcubes after n steps and V_n the total volume, with the initial cube taken to have unit volume ($N_0 = V_0 = 1$).

3.1 Recurrences and closed forms

Each step replaces every retained cube by 20 subcubes, so $N_n = 20\,N_{n-1}$. Each subcube has one-third the side length and hence $1/27$ the volume of its parent; retaining 20 of 27 multiplies the total volume by $20/27$, so $V_n = \frac{20}{27} V_{n-1}$.

Theorem 1. *For all integers $n \geq 0$, $\quad N_n = 20^n \quad$ and $\quad V_n = \left(\dfrac{20}{27}\right)^n$.*

Proof. Both are proved by induction on n.

Claim 1: $N_n = 20^n$. **Basis.** For $n = 0$ the construction is the single starting cube, so $N_0 = 1 = 20^0$. **Step.** Assume $N_k = 20^k$. Each of the N_k retained cubes is subdivided into 27 subcubes of which exactly 20 are kept, and the survivor sets are disjoint across parents, so $N_{k+1} = 20\,N_k = 20 \cdot 20^k = 20^{k+1}$. By induction $N_n = 20^n$ for all $n \geq 0$.

Claim 2: $V_n = (20/27)^n$. **Basis.** $V_0 = 1 = (20/27)^0$. **Step.** Assume $V_k = (20/27)^k$. Passing from step k to $k{+}1$ replaces every retained cube of volume v by 20 subcubes of volume $v/27$, total $20v/27$; summing over all retained cubes multiplies the total volume by $20/27$, so $V_{k+1} = \frac{20}{27} V_k = (20/27)^{k+1}$. By induction $V_n = (20/27)^n$ for all $n \geq 0$. □

Corollary 1. *The Menger sponge has zero volume: since* $0 < 20/27 < 1$, $\lim_{n\to\infty} V_n = \lim_{n\to\infty}(20/27)^n = 0$.

3.2 Surface area: an exact recurrence

Let S_n be the surface area of the level-n solid M_n, with $S_0 = 6$ (the unit cube). We derive a recurrence using self-similarity. The solid M_n consists of 20 copies of M_{n-1}, each scaled by $1/3$; scaling divides area by 9, so, taken in isolation, the 20 copies would contribute $\frac{20}{9}S_{n-1}$. However, wherever two of these copies are face-adjacent, their touching material becomes interior and is hidden from both surfaces, so we must subtract twice the shared area for each adjacency.

Two facts make this precise. First, the number of face-adjacencies among the 20 retained cells is a constant $\alpha = 24$: the retained cells are the 8 corner and 12 edge subcubes of the $3 \times 3 \times 3$ grid, and each corner is face-adjacent to exactly 3 edge subcubes, giving $\alpha = 8 \cdot 3 = 24$ (equivalently, each edge touches 2 corners, $12 \cdot 2 = 24$). Second, the shared face between two adjacent copies carries a Sierpiński carpet, because a face of M_{n-1} is a level-$(n-1)$ carpet; its area on a face of side $1/3$ is $\frac{1}{9}\left(\frac{8}{9}\right)^{n-1}$. Hence

$$S_n = \frac{20}{9}\,S_{n-1} - \frac{2\alpha}{9}\left(\frac{8}{9}\right)^{n-1} = \frac{20}{9}\,S_{n-1} - \frac{48}{9}\left(\frac{8}{9}\right)^{n-1}. \tag{1}$$

Theorem 2. *The surface area of* M_n *is*

$$S_n = 2\left(\frac{20}{9}\right)^n + 4\left(\frac{8}{9}\right)^n,$$

and therefore $S_n \to \infty$ *as* $n \to \infty$.

Proof. Solving the linear recurrence (1): the homogeneous solution is $A(20/9)^n$, and a particular solution of the form $B(8/9)^n$ requires $B(8/9) = \frac{20}{9}B - \frac{2\alpha}{9}$, giving $B = \alpha/6 = 4$. Imposing $S_0 = 6$ fixes $A = 6 - \alpha/6 = 2$, so $S_n = 2(20/9)^n + 4(8/9)^n$. Since $20/9 > 1$, the first term diverges while the second vanishes; hence $S_n \to \infty$, and the ratio $S_n/S_{n-1} \to 20/9 \approx 2.22$. This closed form was verified computationally against a direct exposed-face count for $n = 0, \ldots, 5$ (Table 1). □

Together with the Corollary, this is the three-dimensional analogue of the Koch snowflake's finite-area, infinite-perimeter behavior: the Menger sponge encloses zero volume behind an infinite surface.

3.3 Fractal dimension

For a self-similar set that is the union of M copies of itself each scaled by ratio r, the similarity dimension is $D = \log M / \log(1/r)$. With $M = 20$ and $r = 1/3$,

$$D = \frac{\log 20}{\log 3} \approx 2.7268,$$

strictly between 2 and 3. This agrees with the box-counting dimension, defined as $D = \lim_{\varepsilon\to 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}$, where $N(\varepsilon)$ is the number of boxes of side ε needed to cover the set.

Proposition 1. *At the scales* $\varepsilon = 3^{-m}$*, the covering number is exactly* $N(3^{-m}) = 20^m$*, so the box-counting dimension equals* $\log 20 / \log 3$.

Proof. The level-m approximation M_m is a union of 20^m cubes of side 3^{-m}, and the sponge meets exactly those cubes; hence $N(3^{-m}) = 20^m$ and $\log N / \log(1/\varepsilon) = \log 20^m / \log 3^m = \log 20 / \log 3$ for every m. □

We confirm this empirically in Section 5. (Intermediate scales $\varepsilon \neq 3^{-m}$ exhibit log-periodic oscillation, characteristic of an exactly self-similar set, and are not used for the fit.)

3.4 A base-3 characterization

Membership admits a number-theoretic description generalizing the Cantor set's "no digit 1 in base 3" rule.

Proposition 2. *Write the coordinates* $(x, y, z) \in [0, 1]^3$ *in base* 3*. The point lies in the Menger sponge if and only if, at every digit position,* at most one *of the three coordinates has the digit* 1*.*

Equivalently, a subcube at any level is deleted exactly when at least two of its three base-3 grid coordinates equal the middle value 1. Counting the deleted cells at one level: choosing which two of the three coordinates are the middle gives $\binom{3}{2} = 3$ ways, each with 2 free choices for the third coordinate (0 or 2), yielding $3 \cdot 2 = 6$ face-center cubes, plus the single all-middle center cube, for 7 deletions and $27 - 7 = 20$ survivors, recovering Theorem 1. This rule generates the model in Figure 3.

4 Algorithm and Experiment

4.1 GPU ray marching with a distance estimator

Rendering the sponge by instantiating its $N_n = 20^n$ cubes is infeasible (6.4×10^7 at $n = 6$). Instead we use *ray marching*: for each screen pixel a ray is advanced by the value of a signed-distance function that returns a safe lower bound on the distance to the surface, stopping when it reaches the surface. The distance estimator is itself recursive and mirrors the construction: at each of n folds it maps the point into the repeating $3 \times 3 \times 3$ cell and subtracts the cross-shaped tunnel of the seven deleted cubes. Thus the recurrence of Section 3 reappears as a loop in the shader, with iteration depth n as the loop bound. Because the fold multiplies coordinates by 3^n, which for $n \geq 7$ overruns the available 32-bit floating-point precision, we added a distance-based *level-of-detail* cap that limits the fold depth to the detail resolvable at a given distance, together with a conservative step factor; these remove the precision artifacts and aliasing at high depth. The renderer (Figure 1) is implemented in Three.js/WebGL.

4.2 Box-counting procedure

To estimate the dimension empirically we voxelize M_5 on a 243^3 grid (generated directly from the base-3 rule of Section 3.5) and, for each scale $\varepsilon = 3^{-m}$, count the number of boxes of side 3^{5-m} voxels that contain any occupied voxel. The slope of $\log N(\varepsilon)$ against $\log(1/\varepsilon)$ estimates D.

4.3 Sparse voxel tree

As an alternative representation we build a sparse hierarchical voxel tree. Because the sponge subdivides into $3 \times 3 \times 3 = 27$ cells, its natural hierarchy is a 27-ary tree rather than a standard octree (whose $2 \times 2 \times 2$ split does not align with the base-3 self-similarity): each internal node corresponds to a subcube and stores its 20 occupied children out of the 27, so the tree's branching

structure mirrors the "keep 20 of 27" rule exactly. A node at depth n represents a cube of side 3^{-n}; leaves at the target depth are the retained subcubes. Ray traversal descends only into occupied children, giving an exact intersection with no distance-estimator overshoot.

5 Results

Table 1 lists the exact subcube count, volume, and surface area from Theorems 1 and 2, alongside the feature size 3^{-n}; the surface-area column was reproduced by direct exposed-face counting, confirming the closed form.

n	$N_n = 20^n$	$V_n = (20/27)^n$	$S_n = 2(20/9)^n + 4(8/9)^n$	feature size 3^{-n}
0	1	1.000	6.000	1.000
1	20	0.741	8.000	0.333
2	400	0.549	13.037	0.111
3	8,000	0.406	24.757	0.0370
4	160,000	0.301	51.270	0.0123
5	3,200,000	0.223	110.604	0.00412
6	64,000,000	0.165	240.9	0.00137

Table 1: Count, volume, surface area, and feature size against iteration depth. Note $V_n \to 0$ while $S_n \to \infty$.

Empirical dimension

Figure 2 shows the box count on the 243^3 voxelization. The measured covering numbers are exactly $N(3^{-m}) = 20^m$ for $m = 0, \ldots, 5$, and the least-squares slope is 2.7268, matching the theoretical $\log 20 / \log 3 = 2.7268$ with $R^2 = 1.000$.

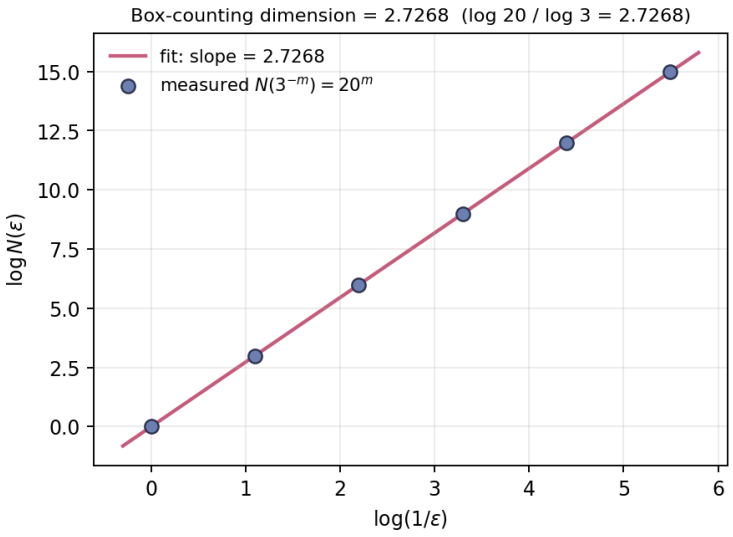


Figure 2: Box-counting dimension. The fitted slope of $\log N(\varepsilon)$ versus $\log(1/\varepsilon)$ is 2.7268, in exact agreement with $\log 20 / \log 3$.

Ray marching versus the voxel tree

The two representations trade off sharply in memory. The distance-estimator renderer stores no geometry, so its memory is $O(1)$ in the depth n, and its cost is per-pixel marching whose accuracy is limited by floating-point precision (mitigated by level-of-detail). The sparse 27-tree stores $\sum_{\ell=0}^{n} 20^{\ell} = (20^{n+1}-1)/19$ nodes, growing as $O(20^n)$, but is exact and free of overshoot artifacts. Table 2 makes the memory growth concrete: the tree is compact and preferable at low-to-moderate depth, while the distance field is the only feasible option for deep or "infinite-zoom" rendering.

depth n	tree nodes $(20^{n+1}-1)/19$	tree memory (4 B/node)	distance-field memory
3	8,421	33.7 KB	$O(1)$
4	168,421	673.7 KB	$O(1)$
5	3,368,421	13.5 MB	$O(1)$
6	67,368,421	269.5 MB	$O(1)$
7	1,347,368,421	5.39 GB	$O(1)$

Table 2: Sparse 27-tree size versus the distance-field renderer. Tree memory grows as $O(20^n)$; the distance field is constant in n.

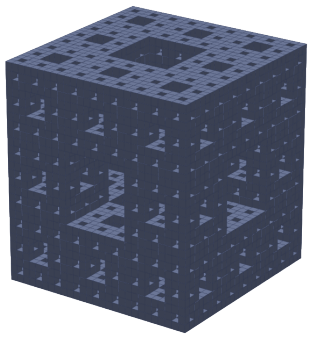


Figure 3: The level-3 sponge ($20^3 = 8000$ cubes) as reconstructed from the sparse voxel tree, generated by the base-3 membership rule.

6 Conclusion

The Menger sponge turns a handful of discrete-mathematical tools into a set of counterintuitive facts. Two first-order recurrences, solved by induction, give the exact count 20^n and volume $(20/27)^n$; a slightly more careful self-similarity argument yields an exact surface-area recurrence whose closed form $2(20/9)^n + 4(8/9)^n$ diverges even as the volume collapses to zero. Self-similarity fixes the dimension at $\log 20/\log 3 \approx 2.7268$, which our box-counting experiment recovered exactly. Building the object two ways sharpened the picture: the recursive distance estimator mirrors the defining recurrence and costs no memory but is precision-limited, while the sparse 27-tree mirrors the "keep 20 of 27" rule as a branching factor and is exact but grows as $O(20^n)$. The same self-similarity that makes the mathematics clean is exactly what the two renderers exploit in opposite ways.

References

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